SAGE-Credit: Adaptive Expert Gating for Credit Default Prediction Under Variable External-Score Availability

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***Abstract*—Credit-default models can deteriorate when highly predictive external-score variables are unavailable at deployment. SAGE-Credit is an availability-aware mixture of CatBoost experts that combines a 929-feature full model with a 922-feature score-free model using an applicant-specific neural gate. Home Credit data for 307,511 applicants were split 80:10:10, and the gate was trained from three-fold out-of-fold expert predictions. On the 30,752-applicant test partition, evaluation covered natural availability, at most two scores, at most one score, and no scores. SAGE-Credit obtained the best mean ROC-AUC (0.7840), average precision (0.2864), Brier score (0.0658), and log loss (0.2376) across the four equally weighted conditions. Paired bootstrapping supported ROC-AUC improvements over the validation-selected individual expert under both partial-score conditions, while differences under natural availability and complete score removal were not supported. The learned gate shifted its mean full-expert weight from 0.901 to 0.033 as scores disappeared, demonstrating adaptive reliance on available information.**

***Index Terms*—credit risk, default prediction, missing data, CatBoost, mixture of experts, model robustness**

I. Introduction

Credit default prediction estimates the probability that borrowers will experience repayment difficulty and is commonly based on application and prior-credit information [1]–[4]. Machine-learning models can exploit nonlinear structure in these data, but performance depends on whether informative variables available during training remain available when predictions are made. Missingness is especially consequential when a small set of variables carries substantial predictive signal [5].

This study focuses on three anonymized Home Credit variables, EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3, plus four summaries derived from them. Their underlying source is undisclosed, so they are treated only as external-score variables rather than as a named commercial credit score. A model trained with these variables may be strongest when they are present, whereas a score-free model can be more stable when they are absent. A fixed blend cannot adapt its reliance to each applicant's information state.

The dataset does not disclose why individual external scores are missing. Accordingly, natural gaps are not assigned to a specific missing-data mechanism, and the controlled removals used here are treated as stress tests rather than as simulations of the process that generated the original missing values [5].

SAGE-Credit applies the mixture-of-experts idea [6] to this availability problem. A learned gate assigns applicant-specific weight to a full expert and a score-free expert. The primary question is whether adaptive weighting can preserve useful score information when present while shifting toward historical and application features as score availability declines. The system is compared with both experts and fixed mixtures under four controlled availability conditions.

II. Methods

A. Data and Feature Construction

The Home Credit Default Risk dataset [7] contains 307,511 labeled loan applications linked to bureau, bureau-balance, previous-application, installment, POS/cash, and credit-card histories. TARGET=1 indicates payment difficulty; 24,825 applicants (8.073%) were positive. Applicants were stratified into fixed 80:10:10 train/validation/test partitions (246,011/30,748/30,752), with all linked records for an applicant confined to one partition.

The six linked historical tables contained 58,133,638 raw records before aggregation, including 27.3 million monthly bureau-status rows. Histories were summarized at applicant level using counts, sums, means, extrema, delinquency/repayment measures, and recent-versus-overall windows. After preprocessing and joining, the tabular representation contained 929 model inputs: 247 application features, 675 engineered historical features, and seven external-score-related features (three raw scores plus four summaries). Numerical application missing values were median-filled; unavailable engineered values were set to zero. A separate 60-month longitudinal representation (60 x 169 per applicant) was used only for preliminary LSTM screening. Feature engineering preserved both long-run and recent behavior. Credit-card and POS histories were summarized over their complete records and over 1-, 3-, 6-, 12-, and 24-month windows; installment windows ranged from 30 to 365 days, previous-application windows from 30 to 730 days, and bureau histories extended to 1,095 days, with bureau-balance status windows through 24 months. Separate aggregates captured balances, account activity, repayment behavior, delinquency, and changes between recent and overall activity. Invalid ratios were converted to

TABLE I
PRELIMINARY MODEL SCREENING ON FIXED TEST PARTITION

Model	ROC-AUC	F1
CatBoost	0.7923	0.3514
XGBoost	0.7914	0.3462
LightGBM	0.7902	0.3457
Logistic regression	0.7745	0.3270
Random forest	0.7734	0.3185
LSTM	0.7508	0.3014

missing values, and the sentinel value 365243 was treated as missing in affected day-based fields before the feature blocks were merged.

For the longitudinal screening branch, each applicant was represented by 60 monthly rows spanning months -60 through -1, creating 18,450,660 applicant-month observations. Monthly credit-card, POS/cash, and bureau-status summaries were combined with selected application variables to form a 307,511 x 60 x 169 tensor. This representation was evaluated only in the LSTM baseline; the final SAGE-Credit system used the richer applicant-level tabular feature set.

B. Baseline Screening and SAGE-Credit

Six candidate models were screened on the fixed applicant split: logistic regression, random forest [9], LightGBM [10], XGBoost [11], and CatBoost [12] on the 929-feature tabular representation, plus an LSTM [13] on the longitudinal representation. CatBoost produced the strongest test ROC-AUC and F1 (Table I), narrowly ahead of the other boosted-tree models, and its screening results informed the choice of modeling family for the final experts.

The screening models were intentionally heterogeneous. Logistic regression used standardized inputs and balanced class weights; the random forest used 500 trees with maximum depth 20. LightGBM allowed 800 boosting rounds at learning rate 0.03 and depth 6, while XGBoost allowed 1,000 rounds at the same learning rate and depth 5 with a positive-class weight of 11.35. The preliminary CatBoost model used depth 4, learning rate 0.03, and up to 4,000 iterations with early stopping, retaining 2,660 trees. The LSTM contained 128 recurrent units followed by 0.3 dropout and a sigmoid output; validation loss stopped training after seven epochs. These experiments were used only to select the modeling family for the final experts.

The full expert used all 929 inputs. The score-free expert excluded the seven external-score-related inputs and used 922 features. An application-only CatBoost model used the 247 application features solely as an ablation. Both primary experts used 2,200 iterations, depth 4, learning rate 0.03, 80% feature sampling per level, L2 leaf regularization 3.0, and random strength 1.0. The SAGE-Credit architecture is shown in Fig. 1.

The gate used 19 standardized inputs: five expert-output features (the two logits, their absolute probability difference, and each expert's distance from 0.5), four score-availability features, and ten historical-coverage features summarizing presence and nonzero engineered information across five linked sources. A 19-to-16 ReLU network with 0.05 dropout and a sigmoid output produced the full-expert weight g; the final probability was p_SAGE = g p_full + (1-g) p_score-free (Fig. 1). The gate was implemented in PyTorch [14] and optimized with AdamW [16] at learning rate 0.003 and weight decay 0.0001.

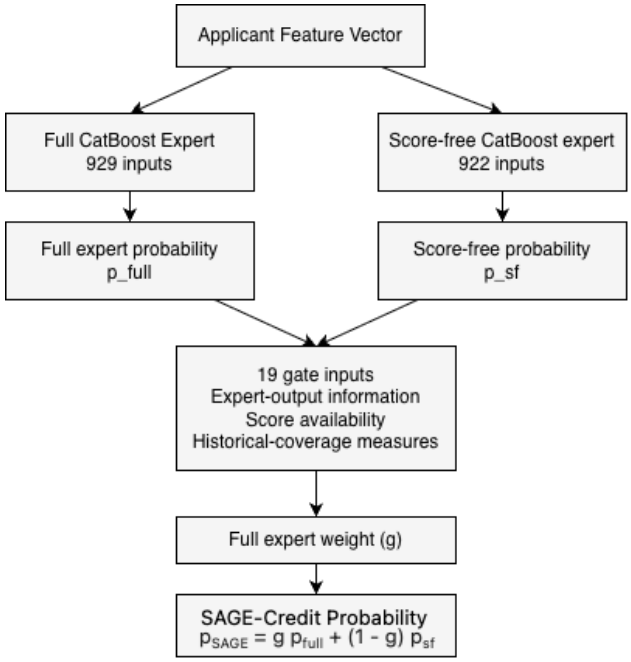


Fig. 1. SAGE-Credit architecture. The gate combines full and score-free CatBoost probabilities using applicant-specific expert, score-availability, and historical-coverage inputs.

TABLE II
FINAL SAGE-CREDIT TRAINING CONFIGURATION

Component	Configuration
Expert inputs	Full: 929; score-free: 922; application-only ablation: 247
CatBoost	2,200 iterations; lr 0.03; depth 4; feature sampling 0.80; L2 3.0; random strength 1.0
OOF gate data	3 stratified folds; 246,011 × 4 = 984,044 rows; each applicant predicted by experts trained on the other folds
Gate network	19 inputs (5 expert, 4 availability, 10 history) → 16 ReLU → 1 sigmoid; dropout 0.05
Optimization	Standardized inputs; AdamW; binary cross-entropy; lr 0.003; weight decay 0.0001; batch 8,192
Epoch selection	Validation loss; max 60; patience 10; min improvement 1e-6; 38 epochs selected
Final gate fit	Reinitialized; all 984,044 OOF rows; 38 epochs
Fixed blend	OOF grid 0.00-1.00 by 0.01; selected full-expert weight 0.48

Three-fold stratified out-of-fold (OOF) predictions [15] across all 246,011 training applicants produced 984,044 gate-training rows across the four availability conditions, preventing the gate from using in-sample expert outputs. Validation selected 38 gate-training epochs; the gate was then reinitialized and fitted on all OOF rows. A global fixed blend with full-expert weight 0.48 was selected on the same OOF predictions (Table II).

Gate-training duration was selected without exposing validation applicants to the optimizer. Final experts fitted only on

the training partition generated validation predictions under all four availability conditions; these validation rows were used solely to monitor loss and select the epoch count. After 38 epochs were selected, the gate was reinitialized and refitted for exactly 38 epochs on the complete 984,044-row OOF training set. No validation or test applicant was used to fit gate parameters.

C. Evaluation

Evaluation used four nested conditions: natural availability, at most two scores, at most one score, and no scores. A fixed seed of 4200 created a deterministic score order for each applicant; the at-most-one condition was nested within the at-most-two condition, and scores already missing remained missing. Derived score summaries and gate availability indicators were recomputed after removal, and models were evaluated without retraining. These conditions are controlled stress tests and are not assumed to reproduce the mechanism that generated naturally missing values [5].

The primary metric was ROC-AUC; average precision (AP), Brier score, and log loss were also reported and computed with scikit-learn [8]. Methods included the full expert, score-free expert, fixed 50/50 mixture, OOF-tuned global blend, and SAGE-Credit. The fixed test partition supplied the reported SAGE-Credit comparisons; it was not used to fit the final experts or gate, although its preliminary screening results had informed the earlier choice of CatBoost. Pairwise ROC-AUC differences were assessed with 5,000 paired applicant bootstrap resamples and percentile 95% confidence intervals [17].

For paired inference, every bootstrap resample used the same sampled applicant positions for both compared methods, preserving within-applicant dependence. For macro comparisons, the same sampled applicants were also carried across all four availability conditions before condition-level ROC-AUC values were averaged. This pairing isolates model differences from variation caused by drawing different test subsets. The validation set determined which individual expert served as the comparator in each condition; the test set then supplied the reported differences and confidence intervals.

III. Results

A. Performance Across Score Availability

Table III summarizes ROC-AUC under each availability condition, and Table IV summarizes mean performance across the four equally weighted conditions. The full expert fell from 0.7921 under natural availability to 0.7508 with no scores, whereas the score-free expert remained at 0.7728 because score removal did not affect its inputs. SAGE-Credit decreased more gradually, from 0.7931 to 0.7726, and achieved the highest four-condition mean ROC-AUC (0.7840). The OOF-tuned fixed blend was strongest at the at-most-one-score condition (0.7822 versus 0.7810 for SAGE-Credit), showing that adaptive gating was not best in every individual regime.

Across the four equally weighted conditions, SAGE-Credit achieved the highest mean AP (0.2864) and the lowest mean Brier score (0.0658) and log loss (0.2376) in addition to the highest mean ROC-AUC (Table IV). Under complete score removal, however, the score-free expert was slightly better than SAGE-Credit on all four metrics; the ROC-AUC difference was only 0.00018. This indicates that the gate approached, but did not exactly equal, the specialized fallback expert when score information vanished.

TABLE III
Test ROC-AUC Across External-Score Availability

Method	Natural	≤2 scores	≤1 score	No scores	Mean
Full expert	0.7921	0.7877	0.7731	0.7508	0.7759
Score-free	0.7728	0.7728	0.7728	**0.7728**	0.7728
50/50 mixture	0.7909	0.7888	0.7821	0.7649	0.7817
OOF-tuned	0.7905	0.7885	**0.7822**	0.7654	0.7817
SAGE-Credit	**0.7931**	**0.7893**	0.7810	0.7726	**0.7840**

TABLE IV
Mean Test Performance Across Four Availability Conditions

Method	ROC-AUC	AP	Brier	Log loss
Full expert	0.7759	0.2745	0.0675	0.2451
Score-free	0.7728	0.2709	0.0666	0.2415
50/50 mixture	0.7817	0.2837	0.0663	0.2404
OOF-tuned	0.7817	0.2836	0.0662	0.2403
SAGE-Credit	**0.7840**	**0.2864**	**0.0658**	**0.2376**

From natural availability to complete score removal, the full expert lost 0.0413 ROC-AUC, whereas SAGE-Credit lost 0.0205. The application-only ablation reached 0.7213 ROC-AUC and 0.1984 AP without scores, compared with 0.7728 ROC-AUC and 0.2709 AP for the score-free expert. The 0.0515 ROC-AUC gap shows that engineered historical features retained substantial predictive value beyond current-application features alone.

The probability-based metrics showed the same robustness pattern as ROC-AUC. From natural availability to complete score removal, SAGE-Credit's AP declined from 0.2998 to 0.2697, its Brier score increased from 0.0651 to 0.0667, and its log loss increased from 0.2344 to 0.2416. Over the same endpoints, the full expert's AP fell from 0.2976 to 0.2391, Brier score worsened from 0.0652 to 0.0725, and log loss from 0.2348 to 0.2674. Under the at-most-one-score condition, the tuned fixed blend retained a slight ROC-AUC/AP edge, but SAGE-Credit produced the lower Brier score (0.066006 versus 0.066047) and lower log loss (0.238629 versus 0.238952), indicating that ranking performance and probability quality did not always order the methods identically.

B. Bootstrap Comparison and Gate Behavior

Using the validation-selected individual expert as comparator, SAGE-Credit's test ROC-AUC differences were +0.0010 under natural availability, +0.0016 with at most two scores, +0.0080 with at most one score, and -0.00018 with no scores. The paired 95% bootstrap intervals excluded zero for both partial-score conditions but included zero under natural availability and complete removal. Thus, the evidence supports an advantage over the selected individual expert specifically when score information was partially available (Fig. 2).

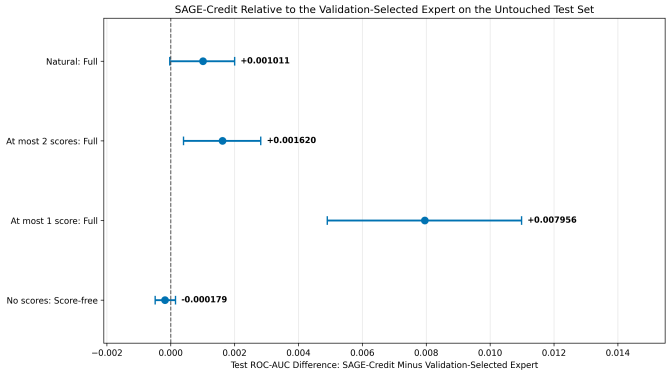


Fig. 2. Paired test ROC-AUC differences between SAGE-Credit and the validation-selected individual expert. Bars show 95% paired-bootstrap intervals; positive values favor SAGE-Credit.

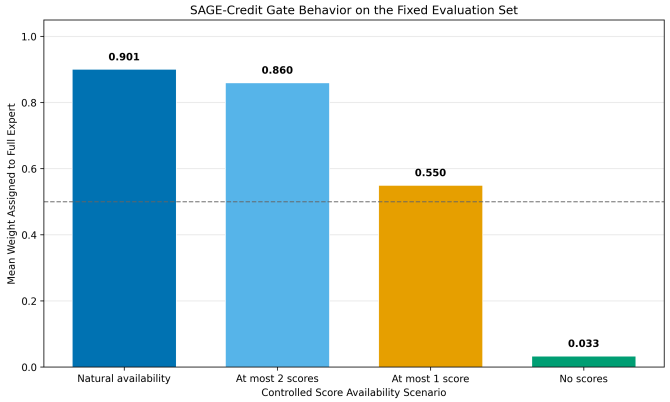


Fig. 3. Mean full-expert gate weight across availability conditions, showing the shift toward the score-free expert as external scores disappear.

The gate shifted monotonically toward the score-free expert as scores disappeared (Fig. 3). Mean full-expert weight fell from 0.901 under natural availability to 0.860 with at most two scores, 0.550 with at most one, and 0.033 with none. Because the weight was produced separately for each applicant, these condition-level means summarize an applicant-specific routing mechanism rather than a fixed scenario rule.

C. Expert Specialization

Feature-importance analysis further supported the experts' specialization. In the full expert, external-score variables and their derived summaries occupied several leading positions, with APP_EXT_SOURCE_MEAN the single most influential feature by CatBoost PredictionValuesChange importance. After the seven score-related inputs were removed, the score-free expert redistributed importance toward application and engineered historical variables. Its leading predictors included APP_ANNUITY_CREDIT_RATIO, BUR_OVERALL_DEBT_CREDIT_RATIO_MAX, DAYS_EMPLOYED, APP_GOODS_CREDIT_RATIO, and BUR_OVERALL_DAYS_CREDIT_MEAN. Historical bureau, installment, and previous-application features therefore remained active predictors rather than serving only as auxiliary inputs.

This change in predictor emphasis complements the application-only ablation. The score-free expert exceeded the application-only expert by 0.0515 ROC-AUC under complete score removal (0.7728 versus 0.7213), indicating that the engineered history supplied a meaningful fallback when external-score information was unavailable. Feature importances are normalized separately within each CatBoost model, so their magnitudes are descriptive of within-model predictor use and should not be compared as absolute effects across experts. They also do not establish causal relationships between any variable and default risk.

IV. Discussion

A. Interpretation and Implications

The preliminary comparison supported CatBoost with the engineered tabular representation as the foundation for SAGE-Credit. In the primary experiment, SAGE-Credit achieved the strongest mean performance across the four predefined availability conditions for ROC-AUC, average precision, Brier score, and log loss. Its advantage was not uniform, however. The gain over the full expert under natural availability was only 0.0010 ROC-AUC and its bootstrap interval contained zero. Under complete score removal, the score-free expert was marginally stronger. The clearest benefit appeared under partial availability, where SAGE-Credit produced bootstrap-supported gains over the validation-selected individual expert.

The gate's behavior helps explain this pattern. When external scores were generally available, it assigned most prediction weight to the full expert, preserving the information carried by the strongest score-related features. As availability declined, it progressively increased reliance on the score-free expert. This adjustment was most useful in the partial-score regimes, where applicants could differ in how much external information remained. When no scores remained, the mean full-expert weight fell to 0.033 and SAGE-Credit converged closely toward the score-free expert rather than continuing to rely on the degraded full model. The gate therefore did not create new predictive information; it redirected each prediction toward the expert whose training inputs better matched the information available for that applicant.

The fixed mixtures provide an important comparison because they also reduce dependence on the full expert. Their limitation is that the mixture weight remains unchanged for every applicant and every availability state. SAGE-Credit instead varied the weight at applicant level using score-availability indicators, historical-coverage measures, and the experts' own predictions. This produced stronger average discrimination and probability accuracy across the controlled regimes, although the OOF-tuned fixed blend remained slightly better on ROC-AUC and average precision in the at-most-one-score condition. The contribution is therefore adaptive robustness across changing information states rather than universal superiority over every simpler alternative.

B. Limitations

Several limitations define the scope of these findings. First, all models were developed and evaluated on one public, anonymized dataset and one fixed train/validation/test split. The paired bootstrap strengthens comparisons within the fixed test partition, but the same partition also informed preliminary model-family selection, so the final estimates are not a fully independent confirmation of that earlier choice. They also do not establish transfer to another lender, loan product, time period, applicant population, or economic environment. Second, external-score availability was altered through controlled removal of seven predefined inputs. This creates a consistent stress test, but naturally occurring missingness may be associated with applicant characteristics, provider coverage, or operational failures. Because the three external variables are anonymized, their real-world sources and the circumstances that make them unavailable cannot be identified from the dataset.

The four availability regimes were also weighted equally when macro performance was reported; those averages therefore summarize robustness across the experimental conditions rather than expected performance under real deployment frequencies. A deterministic router that selects the full or score-free expert solely from score availability was not evaluated. In addition, this study evaluated predictive metrics rather than lending decisions: it did not choose approval thresholds, model unequal financial or social costs of errors, or evaluate subgroup fairness. The CatBoost importance analysis is descriptive, and the study does not claim causal relationships or readiness for operational credit decisions.

C. Future Research

Future work should first test SAGE-Credit on external datasets and with temporal validation, training on earlier applications and evaluating on later ones. Repeating training across additional random seeds and data splits would help determine whether both the performance differences and learned gate behavior remain stable. Availability experiments should also reproduce naturally occurring score-absence patterns when such information is available, or simulate missingness associated with applicant and system characteristics rather than applying the same controlled caps to every applicant.

A second direction is to evaluate calibration within each availability regime and across relevant applicant subgroups, particularly if access to external information differs systematically between groups. Deployment-oriented work would also need cost-sensitive threshold selection, model explanations, monitoring for population and availability drift, and explicit comparison with simpler routing policies. These tests would clarify when a learned gate adds enough value to justify its additional complexity. Such extensions are necessary before operational use, but they build directly on the present result that applicant-specific expert weighting improved robustness in the controlled partial-availability settings evaluated here.

V. Conclusion

SAGE-Credit combined full and score-free credit-default experts with an applicant-specific neural gate. On the 30,752-applicant test partition, the gate progressively shifted toward the score-free expert as external scores disappeared, achieved the strongest mean performance across four controlled availability conditions, and showed bootstrap-supported ROC-AUC gains over the selected individual expert under both partial-score conditions. The findings suggest that adapting model reliance to information availability can improve robustness when predictive inputs are inconsistently accessible.

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